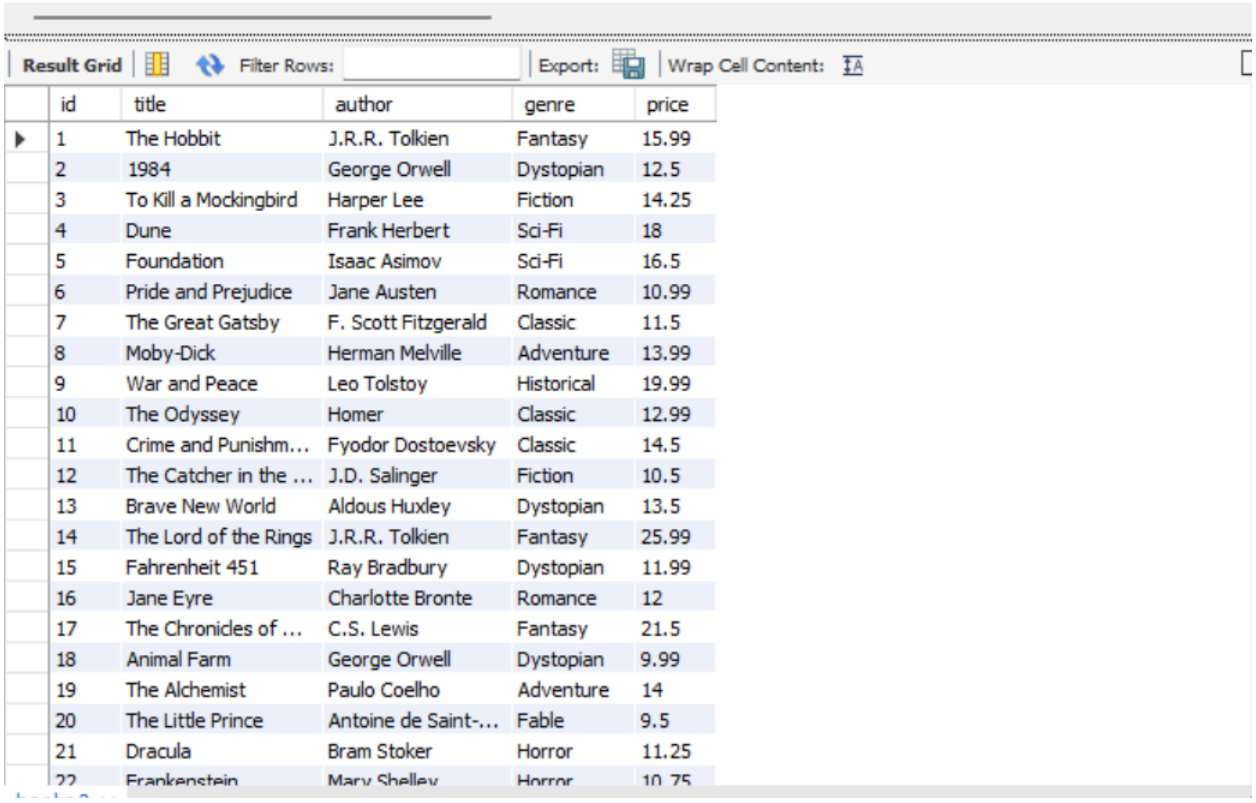
Colin Donahoe

Assignment 5

```
use hw;

select *
from books

```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	title	author	genre	price
1	The Hobbit	J.R.R. Tolkien	Fantasy	15.99
2	1984	George Orwell	Dystopian	12.5
3	To Kill a Mockingbird	Harper Lee	Fiction	14.25
4	Dune	Frank Herbert	Sci-Fi	18
5	Foundation	Isaac Asimov	Sci-Fi	16.5
6	Pride and Prejudice	Jane Austen	Romance	10.99
7	The Great Gatsby	F. Scott Fitzgerald	Classic	11.5
8	Moby-Dick	Herman Melville	Adventure	13.99
9	War and Peace	Leo Tolstoy	Historical	19.99
10	The Odyssey	Homer	Classic	12.99
11	Crime and Punishm...	Fyodor Dostoevsky	Classic	14.5
12	The Catcher in the ...	J.D. Salinger	Fiction	10.5
13	Brave New World	Aldous Huxley	Dystopian	13.5
14	The Lord of the Rings	J.R.R. Tolkien	Fantasy	25.99
15	Fahrenheit 451	Ray Bradbury	Dystopian	11.99
16	Jane Eyre	Charlotte Bronte	Romance	12
17	The Chronicles of ...	C.S. Lewis	Fantasy	21.5
18	Animal Farm	George Orwell	Dystopian	9.99
19	The Alchemist	Paulo Coelho	Adventure	14
20	The Little Prince	Antoine de Saint-...	Fable	9.5
21	Dracula	Bram Stoker	Horror	11.25
22	Frankenstein	Mary Shelley	Horror	10.75

```
// set up an HTTP server off port 3000


const express = require('express');

const app = express();

const PORT = 3000;


// Requires the exported pool object from the local 'mysql.js' file

const pool = require('./mysql');


```

```
//*** This handles the GET request at the root path (/)

app.get("/", async (req, res) => {
    // extract the strings received from the browser
    const category = req.query.category;
    const author = req.query.author;
    const format = req.query.format;

    // Use prepared statements (the '?' placeholder)
    let sql_query = "SELECT id, title, author, genre, price FROM books";
    let params = [];

    if (category) {
        sql_query += " WHERE genre = ?";
        params = [category];
    } else if (author) {
        sql_query += " WHERE author = ?";
        params = [author];
    }

    try {
        // Await the promise from pool.execute. It returns [result,
fields].
        const [result, fields] = await pool.execute(sql_query, params);

        if (format === "text") {
            // start creating the html body for the browser
            let html_body =
"<HTML><STYLE>body{font-family:arial}</STYLE>";
```

```
            html_body += "<BODY>";
            html_body += "<TABLE BORDER=1 WIDTH=800>";

            // print column headings
            html_body += "<TR>";
            // Use fields array from mysql2 for column names
            for (let i = 0; i < fields.length; i++) {
                html_body += (`<TH
style="color:Tomato">${fields[i].name.toUpperCase()}</TH>`);
            }
            html_body += "</TR>";

            // prints rows of table data
            for (let i = 0; i < result.length; i++) {
                const row = result[i];
                html_body += (`<TR>
                    <TD style="vertical-align:top">${row.id}</TD>
                    <TD
style="vertical-align:top"><b>${row.title}</b></TD>
                    <TD style="vertical-align:top">${row.author}</TD>
                    <TD style="vertical-align:top">${row.genre}</TD>
                    <TD
style="vertical-align:top">${row.price.toFixed(2)}</TD>
                </TR>`);
            }

            html_body += "</TABLE>";

            // finish off the html body with a link back to all books
            html_body += "<BR><BR><BR><a
href=http://localhost:3000/?format=text>All Books</a><BR><BR><BR>";
            html_body += "</BODY></HTML>";
```

```
            console.log("Text results generated for query:", req.query);
            res.send(html_body); // send query results back to the browser
        } else {
            // JSON is the default when format is "json" or not provided
            console.log("JSON results generated for query:", req.query);
            res.json(result);
        }

    } catch (err) {
        // Catch and handle database errors cleanly
        console.error("SQL Error in / GET:", err);
        res.status(500).json({ error: 'Database Query Error', details:
err.message });
    }
});



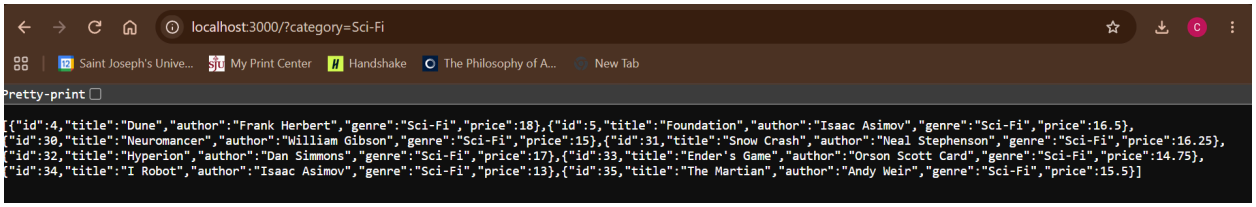
// --- Global Error Handler Middleware ---
// Kept as a fallback for non-route-specific errors or errors passed by
next()
app.use((err, req, res, next) => {
    // Log the error stack for debugging purposes
    console.error(err.stack);
    res.status(500).json({ error: 'An unexpected server error occurred.',
details: err.message });
});

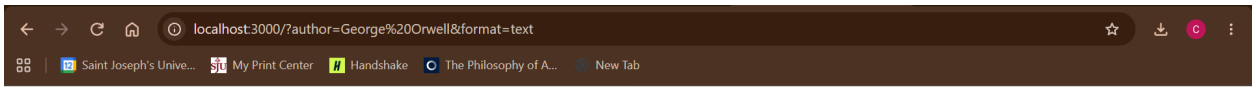


// set up an HTTP server off port 3000
app.listen(PORT, function () {
```

```
    console.log("NodeJS app listening on port " + PORT);

});
```





ID	TITLE	AUTHOR	GENRE	PRICE
2	**1984**	George Orwell	Dystopian	12.5
18	**Animal Farm**	George Orwell	Dystopian	9.99